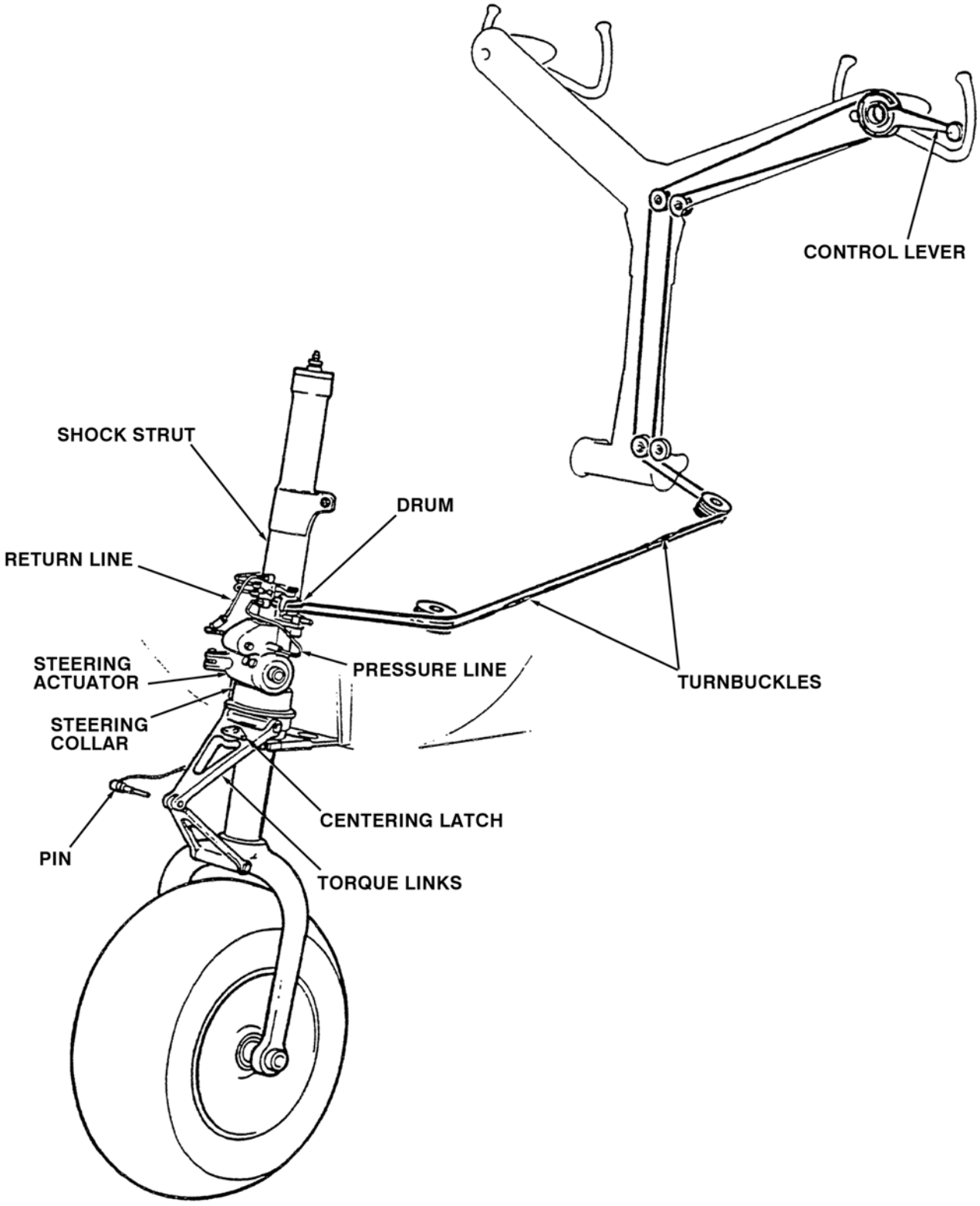
Nose Wheel Steering System – General Data

1. General

Refer to Figure 1 for details.

The nose wheel steering system consists of a hydraulic steering actuator and steering control system. The steering actuator is mounted on the nose gear shock strut and is connected by a link rod to the steering collar and torque links. Hydraulic lines from the steering actuator connect with the main hydraulic system. Steering control comprises a lever and quadrant, which operates a cable and pulley system to rotate a drum and turn a steering valve in the steering actuator. The steering system control lever is located on the pilot's control column. A latch assembly on the upper torque arm will retain the nose wheel in central position in event of hydraulic failure during flight. The latch assembly is disengaged on the ground on compression of nose gear strut.

A luminous tape nose wheel position indicator to indicate the nose wheel neutral reference position. The indicator consists of a luminous tape strip on the steering control lever hub and the control column cover.



Nose Wheel Steering System
Figure 1

2. Servicing

A. Bleed Nose Wheel Steering System

1) Bleeding the nose wheel steering system need only be accomplished if castoring is not smooth, a hydraulic line or component has been replaced, or if the hydraulic system has been dismantled.

2) Bleeding can be carried out with the aircraft on the ground or in the jacked-up position, as follows:

a) Aircraft on ground:

Position aircraft nose wheel on greased plates or similar arrangement. Ensure parking brake is on, or main wheels are securely chocked. Open nose baggage compartment.

Note Two smooth sheets of steel plate with grease applied between them, is a satisfactory platform for nose wheel steering tests. The upper plate should turn smoothly with wheel when steering is operated without any movement between the nose wheel tire and the upper surface of the plate.

b) Aircraft on jacks:

Remove quick-release pin connecting upper and lower torque links on shock strut. Move upper torque link as required, push center latch pin against spring so that latch pin is flush with lower surface and protrudes the upper surface; secure latch pin at upper surface using a hose clamp (jubilee clip). Install quick-release pin at upper and lower torque links. Open nose baggage compartment.

3) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.). Check wing flaps selector corresponds with position of wing flaps.

4) Set POWER SOURCE switch to EXTERNAL, MASTER switch to ON (Pre Mod 6/2314); or set EXTERNAL switch to ON (Post Mod 6/2314). Check electric motor-driven hydraulic pump charges damping accumulator and wheel brakes accumulator, and that motor cuts out when reading on pressure indicators is 1575 ± 50 psi.

5) Operate nose wheel steering fully left and right at least five times.

6) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

7) Check hydraulic reservoir fluid level. (Refer to 12-10-15, Hydraulic System – General Data).

8) If bleeding was carried out with aircraft on jacks, remove hose clamp (jubilee clip) from latch pin at upper torque link. Lower aircraft to ground unless required on jacks for other work. Close nose baggage compartment.

9) If bleeding was carried out with aircraft on the ground, remove greased plates. Close nose baggage compartment.

3. Adjustment/Test

A. Rig Nose Wheel Steering System

1) Rigging of the nose wheel steering system can be accomplished, with the aircraft on the ground or in the jacked-up position, refer to Para 2.A.2) for preparation method. Ensure steering actuator is correctly adjusted (refer to 32-50-11, Nose Wheel Steering Actuator – General Data).

2) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.). Check wing flaps selector corresponds with position of wing flaps.

3) Set POWER SOURCE switch to EXTERNAL, MASTER switch to ON (Pre Mod 6/2314); or set EXTERNAL switch to ON (Post Mod 6/2314). Check electric motor-driven hydraulic pump charges damping accumulator and wheel brakes accumulator, and that motor cuts out when reading on pressure indicators is 1575 ± 50 psi.

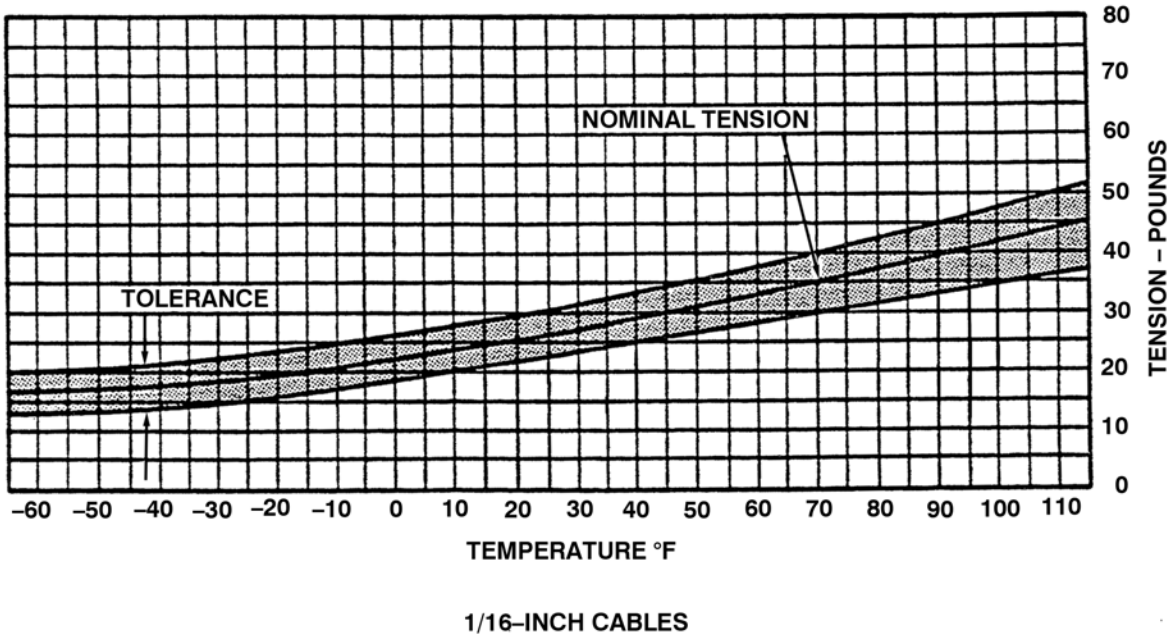
4) Set nose wheel neutral position by turning steering control lever until neutral marks on lever and control column are aligned.

5) Check nose wheel is centered (neutral). If not, adjust turnbuckles to obtain neutral position. Set cable tensions to correct tension given in Figure 2, and lock turnbuckles.

6) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

7) If rigging was carried out with aircraft on jacks, remove hose clamp (jubilee clip) from latch pin at upper torque link. Lower aircraft to ground. Close nose baggage compartment door.

8) If rigging was carried out with aircraft on ground, remove greased plates. Close nose baggage compartment door.



Nose Wheel Steering Cable Tension Graph
Figure 2

B. Function Test Nose Wheel Steering System

1) Testing of the nose wheel steering system can be accomplished with the aircraft on the ground or in the jacked-up position, refer to Para 2.A.2) for preparation method. Ensure steering actuator is correctly adjusted. (Refer to 32-50-11, Nose Wheel Steering Actuator – General Data).

2) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.). Check wing flaps selector corresponds with position of wing flaps.

3) Set POWER SOURCE switch to EXTERNAL, MASTER switch to ON (Pre Mod 6/2314); or set EXTERNAL switch to ON (Post Mod 6/2314). Check electric motor-driven hydraulic pump charges damping accumulator and wheel brakes accumulator, and that motor cuts out when reading on pressure indicators is 1575 ± 50 psi.

4) Steer nose wheel left and right. From straight ahead position to full left or full right. Check operation time is 5.0 to 7.0 seconds.

5) Steer nose wheel to any intermediate position and apply torque manually to the lower leg assembly. Check there is no movement of the wheel from the selected position.

6) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

7) Discharge hydraulic system pressure by operating wing flaps.

8) Apply torque manually to the lower leg assembly. Check that there is a steady resisting force throughout the entire steering range.

9) If function test was carried out with aircraft on jacks, remove hose clamp (jubilee clip) from latch pin at upper torque link. Lower aircraft to ground.

10) If function test was carried out with aircraft on ground, remove greased plates.

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